

Connect 4 Agent

Sistemas Inteligentes 2

Practical Work nº1

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1 Introduction

Introduction here Xana

2 Server Message Format

Mensagens do tipo:

```
{'type': 'state', 'board': [[0, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 0],
    [0, 0, 0, 0, 2, 0, 0], [0, 0, 0, 0, 1, 0, 0], [0, 0, 0, 0, 2, 0, 0],
    [1, 0, 1, 0, 2, 0, 0]], 'valid_actions': [0, 1, 2, 3, 4, 5, 6], '
    current_turn': 1}
```

Estado de jogo convertido para Flattened Tuple, Zobrist Hashing, Bitboards

2.1 Flattened Tuple

(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 2, 0, 0, 1, 0, 1, 0, 2, 0, 0)

2.2 Bitboards

This represents the board using two large integers, one for each player. Each bit in the integer corresponds to a square on the grid ($index = row \times 7 + column$)

Player 1: 171832246272

Player 2: 554051043328

If Player 1 has a piece at Row 5, Col 0, you set the 35th bit (2^{35}). By summing these powers of 2 (using bitwise OR), you get a single number that perfectly describes all of one player's pieces. This is the fastest way to check for wins using bitwise masks.

171832246272 representa em decimal o número binário:

[illegible]

$7 \times 6 = 42$ squares = 42 bits (usamos 64 bits, sobrando 22 bits para outras informações ou para evitar overflow)

2.3 Zobrist Hashing

This is the standard for Transposition Tables. It converts the entire board into a single unique 64-bit number using the XOR operator.

Zobrist Hash: 11436696887795538186

(Example based on a seeded random table)

O melhor para verificar vitórias é o Bitboards

O melhor para guardar é o Zobrist Hashing